

LAB-3

Q Implement A* search algorithm

Initial state

5	4	
6	1	8
7	3	2

$h=8$

$g=0$

$f=8$

Goal state

	1	2
3	4	5
6	7	8

$g=1$

$h=8$

$f=9$

5		4
6	1	8
7	3	2

5	4	8
6	1	
7	3	2

$g=1$

$h=8$

$f=9$

$g=2$

$h=7$

$f=9$

5	1	4
6		8
7	3	2

5	4	
6	1	8
7	3	2

	4	5
6	1	8
7	3	2

5	1	4
	6	8
7	3	2

$g=3$

$h=7$

$f=10$

5	1	4
6	3	8
7		2

$g=3$

$h=7$

$f=10$

6	4	5
	1	8
7	3	2

$g=3$

$h=$

$f=$

5	1	4
6	8	
7	3	2

$g=3$

$h=7$

$f=10$

5	1	4
6	3	8
7	2	

$g=4$

$h=6$

$f=10$

5	1	4
6	3	8
7	2	

$g=4$

$h=7$

$f=11$

5	1	4
	3	8
6	7	2

$g=5$

$h=5$

$f=10$

5	1	4
6	3	8
7		2

$h=6$

$g=5$

$f=11$

5	1	4
3		8
6	7	2

$g=6$

$h=4$

$f=10$

Algorithm:

① Initialize

- create start node with initial state and set its cost to 0.
- create an empty priority queue called frontier and add the start node to it.
- create an empty set called explored to store visited states

② while frontier is not empty

- remove the node with the lowest cost from the frontier
- if the current node's state matches the goal state, return path to this node
- Add the current node's state to the explored

③ for each neighbour of the current node

- calculate the depth g as the current node depth + 1
- calculate the heuristic h by counting the no. of misplaced tiles of the neighbour's state
- calculate the total cost

$$f(n) = g(n) + h(n)$$

- If the neighbour's state is not in explored

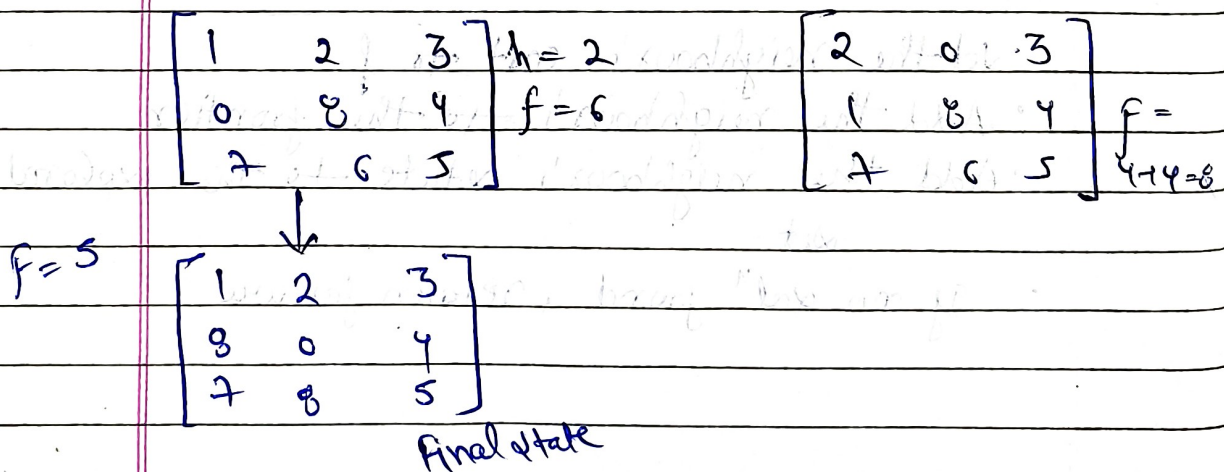
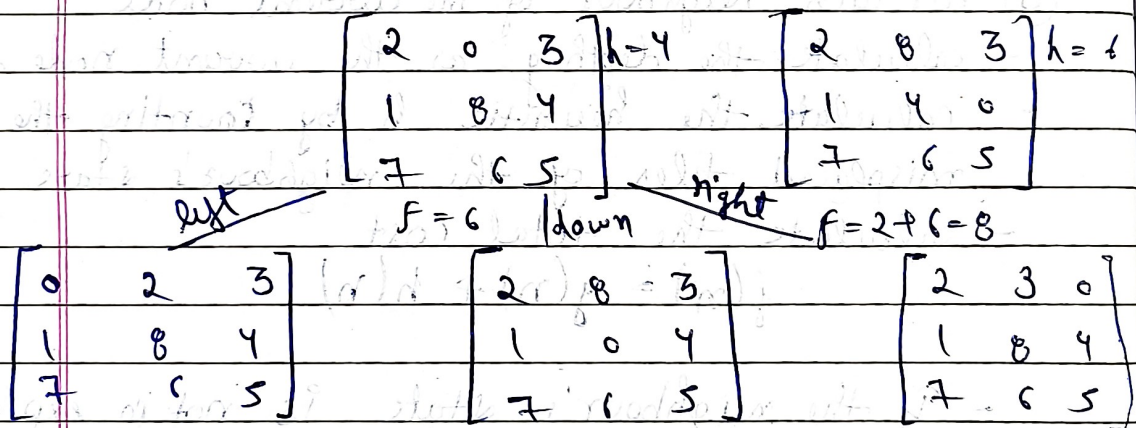
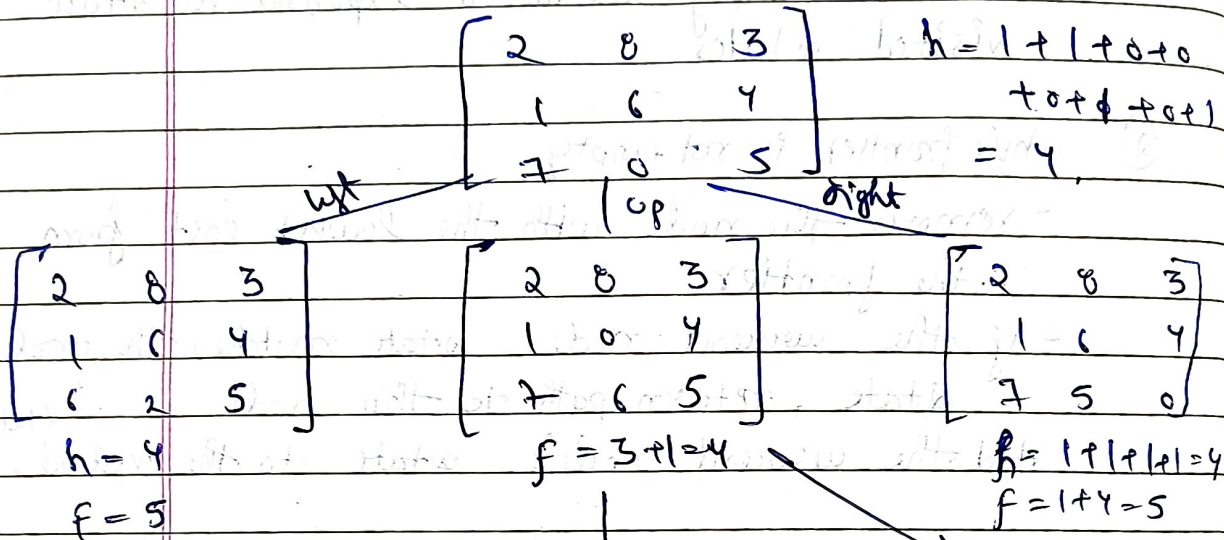
- set the neighbour's cost to f
- Add the neighbour to the frontier
- Add the neighbour's state to the explored set

- If no solⁿ found, return failure

X Manhattan

Final state

1 2 3
8 0 4
7 6 5



Algorithm:

(i) Initialize

- start with initial state and goal state

for each state, calculate

$g(n) \rightarrow$ depth of the node

$h(n) \rightarrow$ sum of manhattan distance of all tiles

compute $f(n) = g(n) + h(n)$

- Add the state to the open list and track visited states
- select the state with the lowest $f(n)$ and generate its neighbors by swapping the blank tile with neighbouring tiles
- continue until the goal state is reached or the open list is empty
- for each neighbour of the current node
 - calculate the depth of the current node's depth + 1
 - calculate the heuristic h using the manhattan distance for the neighbouring state
 - calculate the total cost $f = g + h$
- If the neighbour's state is not in explored,
 - set the neighbour cost to f
 - Add the neighbour to the frontier
 - Add the neighbour's state to the explored^{set}
- If no solution found, return failure